# Beyond Black–Scholes: A Computational Study of Option Pricing under Stochastic Volatility and Jumps

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Companion paper to the `options-pricing-engine` repository

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**Abstract**

The Black–Scholes–Merton model remains the cornerstone of option pricing, yet its assumption of constant volatility is flatly contradicted by the *volatility smile* observed in traded option markets. This paper documents a self-contained pricing library that implements and cross-validates a hierarchy of models and numerical methods: the Black–Scholes benchmark with analytic Greeks; lattice, Monte Carlo, and quasi-Monte Carlo numerical schemes; the Heston stochastic-volatility model priced by a semi-analytic Fourier integral; the Merton jump-diffusion model; path-dependent (Asian, barrier, lookback) options; and American options via Least-Squares Monte Carlo. Every method is validated against an independent benchmark—Fourier prices against Monte Carlo, lattice prices against closed forms, and simulation estimators against analytic control variates. We apply the library to real S&P 500 (SPY) option quotes and show that the observed implied-volatility skew, which Black–Scholes cannot represent, is reproduced by calibrating the Heston and Merton models to the market: a vega-weighted least-squares fit attains 0.60% and 1.22% implied-volatility RMSE respectively across 903 liquid quotes spanning six maturities. We further demonstrate that a geometric-average control variate reduces Asian-option simulation variance by a factor of roughly 36×, and that a scrambled Sobol sequence improves the European-option convergence rate from $O(n^{-1/2})$ toward $O(n^{-1})$. All inputs are real market data and all results are reproducible from the accompanying open-source code.

## 1 Introduction

The 1973 Black–Scholes–Merton (BSM) framework (Black & Scholes, 1973; Merton, 1973) gives a closed-form price for European options under the assumption that the underlying asset follows a geometric Brownian motion with constant volatility. Its elegance and tractability made it the lingua franca of derivatives markets. However, two robust empirical facts contradict its assumptions. First, implied volatilities backed out of market prices are not constant across strikes: out-of-the-money equity puts trade at systematically higher implied volatilities than at-the-money options, producing the characteristic *volatility skew*. Second, asset returns exhibit heavy tails and sudden discontinuities that a pure diffusion cannot generate.

Two families of models address these deficiencies. *Stochastic-volatility* models, of which the Heston (Heston, 1993) model is the canonical example, let the instantaneous variance follow its own mean-reverting diffusion correlated with the asset. *Jump-diffusion* models, introduced by Merton (1976), superimpose a compound Poisson process of discontinuous jumps on the diffusion. Both generate non-constant implied-volatility curves and richer return distributions.

This paper accompanies an open-source pricing library and serves two purposes. Pedagogically, it derives and connects the main pricing techniques used on a modern derivatives desk. Empirically, it demonstrates—with reproducible numerical experiments—how the advanced models cure the shortcomings of Black–Scholes, and how careful numerical design (variance reduction, low-discrepancy sequences, regression-based early-exercise rules) makes these models computationally practical. The guiding methodological principle is *cross-validation*: wherever a quantity can be computed two independent ways, we compute it both ways and verify agreement.

The remainder of the paper is organised as follows. Section 2 reviews the Black–Scholes benchmark and its Greeks. Section 3 covers the lattice, Monte Carlo, and quasi-Monte Carlo methods used throughout. Section 4 introduces the Heston and Merton models and shows that they reproduce the smile. Section 5 treats path-dependent and American options. Section 6 calibrates the Heston and Merton models to real S&P 500 option data. Section 7 collects the numerical results and Section 8 concludes.

## 2 The Black–Scholes Benchmark

Under the risk-neutral measure $\mathbb{Q}$, the underlying follows

$$\mathrm{d}S_t = (r-q)S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t, \tag{1}$$

where $r$ is the risk-free rate, $q$ the continuous dividend yield, and $\sigma$ the (constant) volatility. The time-0 price of a European call with strike $K$ and maturity $T$ is

$$C = S_0 e^{-qT}\Phi(d_1) - Ke^{-rT}\Phi(d_2), \qquad d_{1,2} = \frac{\ln(S_0/K) + (r-q\pm\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \tag{2}$$

with $\Phi$ the standard normal CDF and the put price following from put–call parity, $C - P = S_0e^{-qT} - Ke^{-rT}$.

**Greeks.** The library computes the analytic sensitivities $\Delta = \partial C/\partial S$, $\Gamma = \partial^2 C/\partial S^2$, $\mathcal{V} = \partial C/\partial\sigma$ (vega), $\Theta = \partial C/\partial t$, and $\rho = \partial C/\partial r$ in closed form. These serve both for hedging and as a consistency check: e.g. $\Gamma$ is identical for calls and puts, and $0 < \Delta_{\text{call}} < 1$ while $-1 < \Delta_{\text{put}} < 0$, both verified in the test suite. Figure 1 plots the principal Greeks against spot: the call delta rises sigmoidally from 0 to 1 as the option moves into the money, while gamma and vega—the second-order and volatility sensitivities—peak near the money, where optionality is greatest, and decay in both tails.

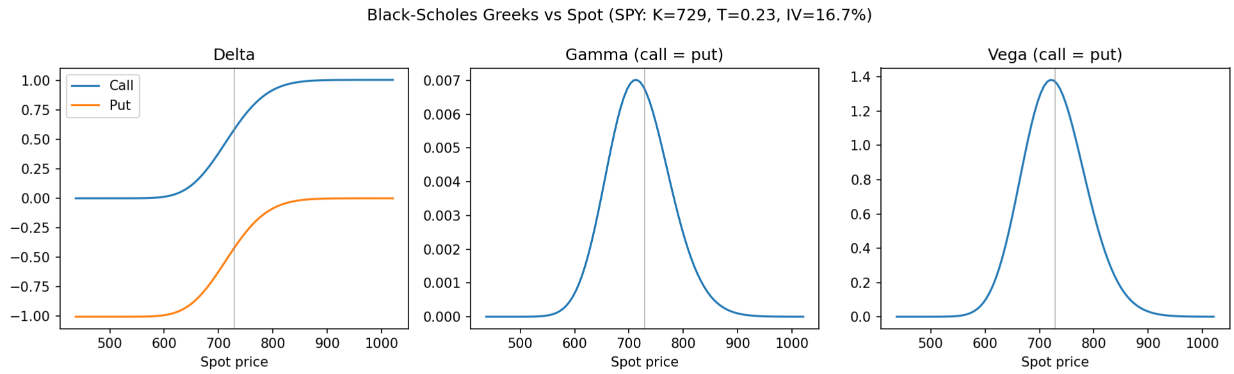



Figure 1: Black–Scholes Greeks as a function of spot ($K = 100$, $T = 1$, $\sigma = 20\%$). Delta is bounded in $(0, 1)$ for the call and $(-1, 0)$ for the put; gamma and vega are identical for calls and puts and are maximal near the money.

**Implied volatility.** Given a market price, the implied volatility is the $\sigma$ solving (2). We invert it by Newton–Raphson using the analytic vega, falling back to Brent's method on the rare occasions vega is too small for stable Newton steps (deep out-of-the-money, short maturity). The implied-volatility map is the lens through which all later models are compared: each model's prices are converted to BSM-implied volatilities and plotted against strike.

# 3 Numerical Methods

## 3.1 Binomial lattice

The Cox–Ross–Rubinstein (CRR) tree (Cox et al., 1979) discretises $[0, T]$ into $N$ steps with up/down factors $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$ and risk-neutral probability $p = (e^{(r-q)\Delta t} - d)/(u - d)$. The option value is computed by backward induction. The lattice converges to (2) as $N \to \infty$ at rate $O(1/N)$ (with the familiar even/odd oscillation), confirmed in Figure 7a. Its practical importance is that it handles *American* exercise directly by taking the maximum of continuation and intrinsic value at every node—something the closed form cannot do.

## 3.2 Monte Carlo and variance reduction

Monte Carlo estimates the discounted risk-neutral expectation $e^{-rT}\mathbb{E}^{\mathbb{Q}}[\,\text{payoff}\,]$ by simulation. Its error decays as $O(n^{-1/2})$ in the number of paths $n$, so variance reduction is essential. We implement two standard techniques:

- **Antithetic variates**: pairing each normal draw $Z$ with $-Z$ removes sampling asymmetry and halves the effective variance for monotone payoffs.
- **Control variates**: subtracting a correlated quantity with known expectation. For a European call the discounted terminal spot is an exact control ($\mathbb{E}[e^{-rT}S_T] = S_0 e^{-qT}$); for an arithmetic Asian option the *geometric* Asian option—which has a closed form—is a near-perfect control (Section 5).

### 3.3 Quasi-Monte Carlo

Replacing pseudo-random draws with a low-discrepancy *Sobol* sequence (with Owen scrambling, inverted through $\Phi^{-1}$) accelerates convergence for smooth, low-dimensional integrands. For the one-dimensional European payoff we observe (Figure 8b) the error decaying close to $O(n^{-1})$ rather than $O(n^{-1/2})$—two orders of magnitude fewer samples for the same accuracy at $n \sim 10^4$.

## 4 Beyond Black–Scholes

### 4.1 The Heston stochastic-volatility model

Heston (Heston, 1993) models the variance $v_t$ as a Cox–Ingersoll–Ross diffusion correlated with the asset:

$$\mathrm{d}S_t = (r-q)S_t\,\mathrm{d}t + \sqrt{v_t}\,S_t\,\mathrm{d}W_t^S, \tag{3}$$

$$\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}\langle W^S, W^v\rangle_t = \rho\,\mathrm{d}t, \tag{4}$$

with mean-reversion speed $\kappa$, long-run variance $\theta$, vol-of-vol $\xi$, and leverage correlation $\rho$ (typically negative for equities). The parameter $\rho$ controls the skew and $\xi$ the convexity of the smile; the Feller condition $2\kappa\theta > \xi^2$ guarantees $v_t > 0$.

Heston's tractability comes from the closed-form characteristic function of the log-price $x_T = \ln S_T$. Writing $\phi(u) = \mathbb{E}[e^{iux_T}]$, the price admits the Fourier representation

$$C = S_0 e^{-qT} P_1 - K e^{-rT} P_2, \qquad P_j = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-iu\ln K}\,\phi_j(u)}{iu}\right]\mathrm{d}u, \tag{5}$$

where $\phi_2 = \phi$ and $\phi_1(u) = \phi(u-i)/\phi(-i)$, and $\phi(-i) = S_0 e^{(r-q)T}$. We evaluate (5) with adaptive quadrature, using the Albrecher et al. (Albrecher et al., 2007) "good" formulation of $\phi$ that avoids the branch-cut discontinuity (the "Little Heston Trap") and is stable at long maturities. As an independent check, a full-truncation Euler Monte Carlo (Lord et al., 2010) reprices the same options; the two agree to within Monte Carlo error plus a small discretisation bias (Section 7).

### 4.2 The Merton jump-diffusion model

Merton (Merton, 1976) adds log-normal jumps arriving as a Poisson process of intensity $\lambda$:

$$\frac{\mathrm{d}S_t}{S_{t^-}} = (r - q - \lambda k)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + (J-1)\,\mathrm{d}N_t, \qquad \ln J \sim \mathcal{N}(\mu_J, \sigma_J^2), \tag{6}$$

with compensator $k = \mathbb{E}[J-1] = e^{\mu_J + \sigma_J^2/2} - 1$ ensuring the discounted price is a martingale. Conditioning on the number of jumps yields Merton's series, a Poisson-weighted sum of Black–Scholes prices,

$$C = \sum_{n=0}^{\infty} \frac{e^{-\lambda' T}(\lambda' T)^n}{n!}\, C_{\mathrm{BS}}(S_0, K, T, r_n, \sigma_n)\,, \tag{7}$$

where $\lambda' = \lambda(1+k)$, $\sigma_n^2 = \sigma^2 + n\sigma_J^2/T$, and $r_n = r - \lambda k + n(\mu_J + \frac{1}{2}\sigma_J^2)/T$. The use of the jump-adjusted intensity $\lambda'$ (rather than $\lambda$) in the Poisson weights is precisely what makes the per-term discounting consistent; an independent terminal-value Monte Carlo confirms the series (Section 7).

### 4.3 Reproducing the volatility smile

Both models break the constant-volatility straitjacket: pricing options across strikes and inverting to BSM-implied volatility yields a non-flat curve. The calibrated-Heston implied-volatility surface (Figure 8a) shows a pronounced downward skew, steepest at short maturities and flattening as $T$ grows, matching the documented term structure of equity-index skew. Section 6 confirms this directly against real S&P 500 quotes and reports the calibration quality.

The mechanism behind the smile is visible in the risk-neutral return distribution. Figure 2 compares the simulated terminal log-returns under the three models, all parameterised by the SPY calibration of Section 6. Black–Scholes is Gaussian by construction; Heston, through its negative leverage correlation, is left-skewed, placing more mass on large downward moves and thereby richening out-of-the-money puts; Merton's jumps produce a sharply peaked body with heavy tails. It is precisely these departures from normality—skewness and excess kurtosis—that translate, through the inversion of (2), into the observed implied-volatility skew.

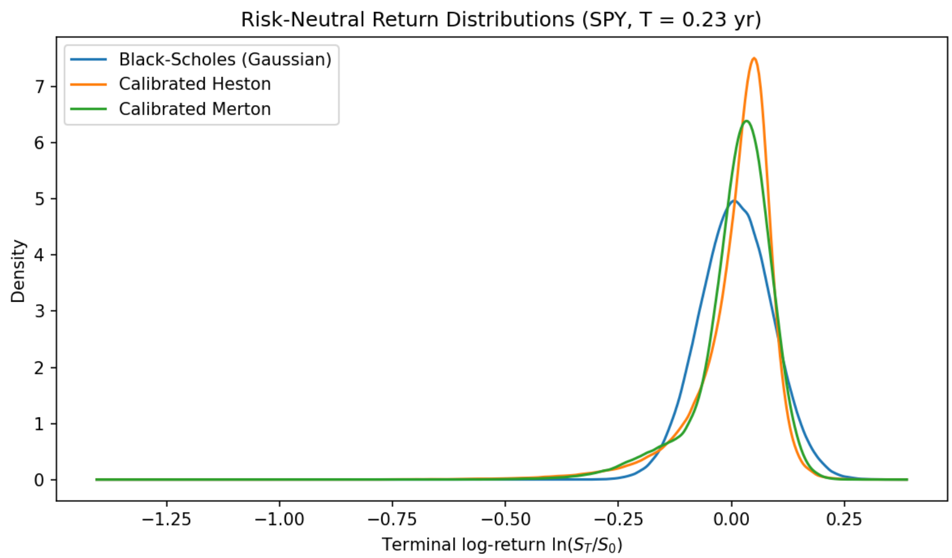



Figure 2: Risk-neutral densities of the terminal log-return $\ln(S_T/S_0)$ at the SPY reference maturity ($T = 0.23$), using the calibrated parameters. Heston is left-skewed and Merton fat-tailed relative to the Gaussian Black–Scholes benchmark; these shape deviations are the source of the skew.

The two Heston parameters that shape the smile play distinct, interpretable roles, illustrated in Figure 3. The leverage correlation $\rho$ tilts the smile: $\rho < 0$ produces the downward equity skew, $\rho > 0$ an upward skew, and $\rho = 0$ a symmetric smile. The vol-of-vol $\xi$ governs convexity: raising $\xi$ fattens both tails and deepens the curvature of the smile, while leaving the at-the-money level largely unchanged. This decoupling is what makes the model practically calibratable—$\rho$ and $\xi$ target the slope and curvature of the observed surface almost independently.

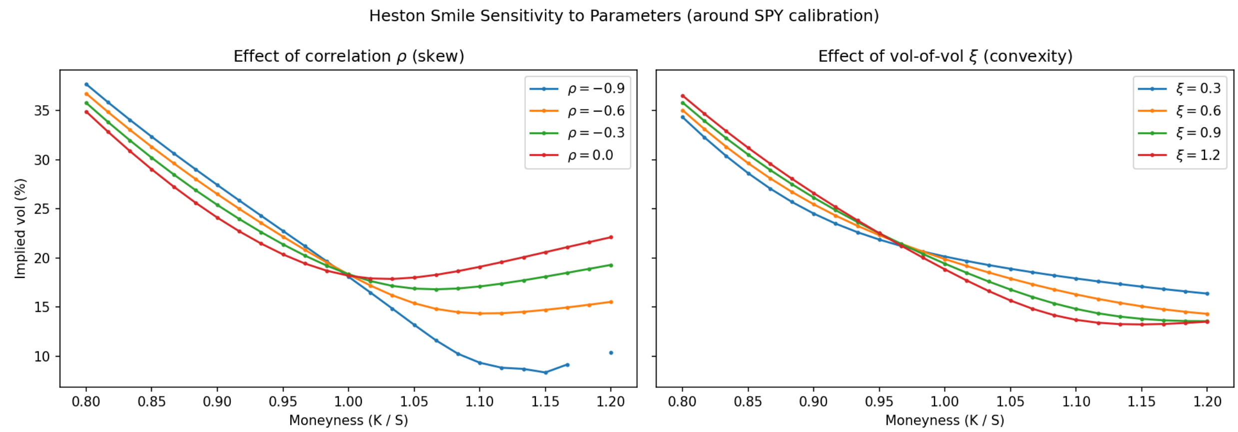



Figure 3: Heston smile sensitivity. Left: the correlation $\rho$ controls the sign and steepness of the skew. Right: the vol-of-vol $\xi$ controls the convexity of the smile.

# 5 Path-Dependent and American Options

## 5.1 Exotics by Monte Carlo

Options whose payoff depends on the whole price path generally lack closed forms, making simulation the natural tool.

**Asian options.** For an arithmetic-average Asian option no closed form exists, but the *geometric*-average version does—the geometric mean of log-normals is log-normal—giving a Black–Scholes-type formula. We use it as a control variate: the arithmetic and geometric payoffs are over 0.99 correlated, so subtracting the geometric control (with its analytic mean) cuts the standard error by roughly 36× at fixed sample size (Section 7).

**Barrier options.** Knock-in and knock-out options are priced by monitoring simulated paths against the barrier. We validate the estimator through *in–out parity*: a knock-in plus the corresponding knock-out must equal the vanilla option, since on every path exactly one of the two pays. The simulated sum matches the analytic vanilla price to within Monte Carlo error.

**Lookback options.** Floating-strike lookbacks pay relative to the path's running extremum and, as expected, are worth strictly more than the corresponding vanilla option—a monotonicity the test suite enforces.

## 5.2 American options by Least-Squares Monte Carlo

American options permit early exercise, requiring the holder to compare immediate intrinsic value against the (unknown) continuation value at each date. The Longstaff–Schwartz algorithm (Longstaff & Schwartz, 2001) estimates the continuation value by least-squares regression of discounted future cash flows on a polynomial basis of the current spot, restricted to in-the-money paths, then applies the resulting exercise policy by backward induction. For an American put our LSM estimate agrees with a high-resolution binomial lattice to within simulation error (Section 7), confirming both implementations.

The early-exercise decision is summarised by the *exercise boundary* $S^*(t)$: the critical spot below which (for a put) immediate exercise dominates holding. Figure 4 extracts this boundary from the lattice. It lies strictly below the strike, rises monotonically toward $K$ as maturity approaches—reflecting the shrinking time value that early exercise forgoes—and meets the strike at expiry. The region beneath the curve is where the American option is worth more dead than alive, the precise locus that distinguishes it from its European counterpart.

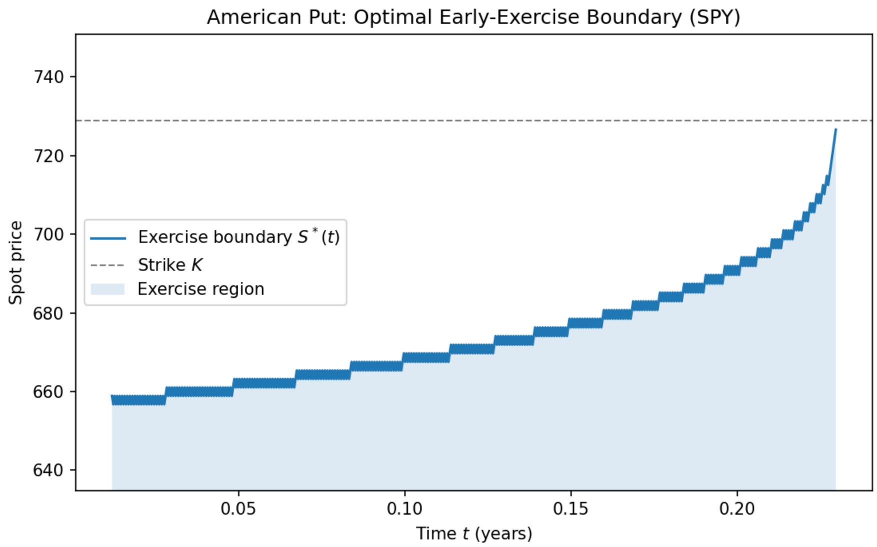



Figure 4: Optimal early-exercise boundary $S^*(t)$ for an American put on the SPY reference contract ($K = 729$, $T = 0.23$, $\sigma = 16.7\%$), extracted from the binomial lattice. Exercise is optimal in the shaded region below the boundary; the boundary rises to the strike at expiry. The staircase pattern reflects the discrete lattice nodes.

# 6 Empirical Application: Calibration to S&P 500 Options

## 6.1 Data

We take the full option chain on the SPDR S&P 500 ETF (SPY) from Yahoo Finance, recorded with the spot at \$728.99 (close of 2026-06-26). The risk-free rate $r = 3.66\%$ is the 13-week US Treasury-bill yield; for each expiry the forward $F$ is implied directly from put-call parity, which fixes the carry $q = r - \ln(F/S)/T$ without a separate dividend assumption and makes the put and call wings consistent at the money. We retain liquid out-of-the-money quotes only (positive bid/ask, relative spread below 20%, open interest $\geq 25$, moneyness in $[0.85, 1.15]$): puts below the forward and calls above it. This yields **903 quotes across six maturities** from 21 to 126 days. The resulting implied-volatility skew (Figure 5) is the classic equity-index shape: out-of-the-money puts trade 10–20 volatility points above out-of-the-money calls, and the skew is steepest at the shortest maturity and flattens with $T$.

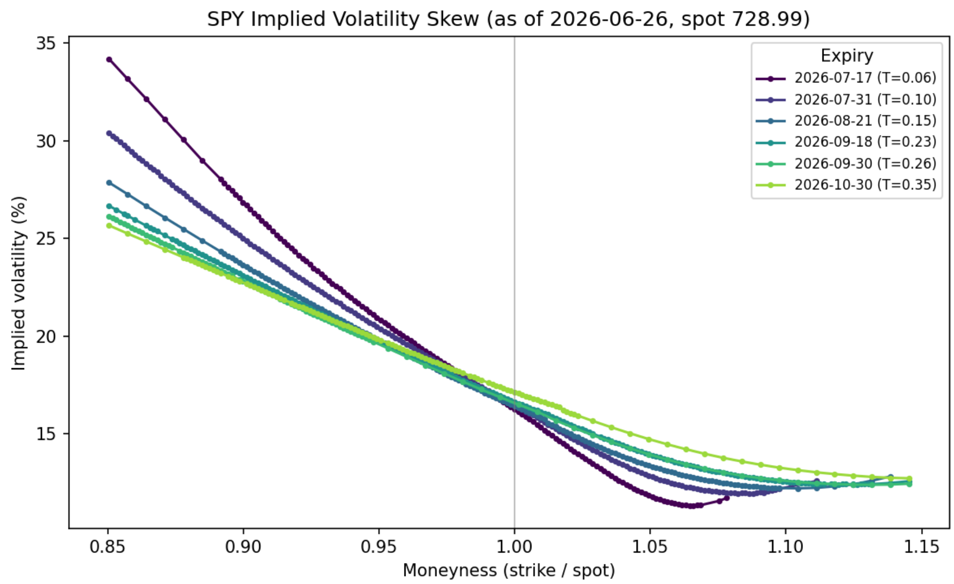



Figure 5: Market implied-volatility skew of SPY options (as of 2026-06-26, spot \$728.99), one curve per expiry. The pronounced negative skew and its flattening with maturity are exactly the features Black–Scholes cannot represent.

## 6.2 Calibration

Calibration is the inverse problem of pricing: given the market quotes, find the model parameters that reproduce them. We minimise the vega-weighted squared pricing error

$$\min_{\Theta} \sum_i w_i^2 \left( C^{\text{model}}(K_i, T_i; \Theta) - C_i^{\text{mkt}} \right)^2, \qquad w_i = 1/\mathcal{V}_i, \tag{8}$$

by the Trust-Region Reflective least-squares method with parameter bounds. The vega weighting $w_i = 1/\mathcal{V}_i$ converts each dollar residual into an approximate implied-volatility residual, so the cheap out-of-the-money options that define the skew are not swamped by expensive at-the-money ones. We calibrate the five Heston parameters $(v_0, \kappa, \theta, \xi, \rho)$ and the four Merton parameters $(\sigma, \lambda, \mu_J, \sigma_J)$ to the same 903 quotes; the resulting fits are shown in Figure 6 and Table 1.

Table 1: Calibrated parameters and fit quality on the 903 SPY quotes. "Vol RMSE" is the root-mean-square error in Black–Scholes implied-volatility points.

| **Heston** | | **Vol RMSE** $= 0.60\%$ |
|---|---|---|
| $v_0 = 0.0259$ | $\kappa = 8.89$ | $\theta = 0.0416$ |
| $\xi = 1.48$ | $\rho = -0.71$ | (Feller $2\kappa\theta > \xi^2$ violated) |
| **Merton** | | **Vol RMSE** $= 1.22\%$ |
| $\sigma = 0.111$ | $\lambda = 0.85\,\text{yr}^{-1}$ | |
| $\mu_J = -0.130$ | $\sigma_J = 0.085$ | |

The economic content of the estimates is exactly what the skew implies. Heston's correlation $\rho = -0.71$ is strongly negative—the leverage effect that tilts the smile downward—and Merton's mean jump $\mu_J = -0.13$ with intensity $\lambda = 0.85\,\mathrm{yr}^{-1}$ encodes the market's pricing of sudden downward moves (crash risk). Heston fits the surface markedly better than Merton (0.60% vs. 1.22% vol RMSE): as Figure 6 shows, its two skew-shaping parameters track the smile across maturities, whereas Merton's fit deteriorates in the wings and at the shortest maturity, where even Heston cannot fully match the steepness—a documented limitation of diffusive stochastic-volatility models at short horizons. The calibrated $\kappa$ is the least well-identified parameter, a known near-degeneracy with $\xi$ and $\theta$.

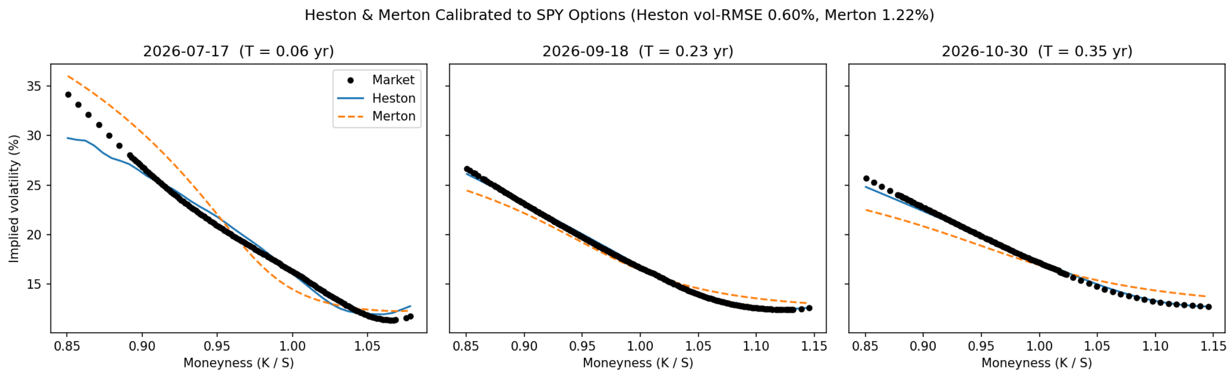



Figure 6: Heston and Merton calibrated to SPY options at three maturities. Markers are market implied volatilities; lines are the calibrated models. Heston (solid) fits the skew tightly except at the extreme short-dated wings; Merton (dashed) captures the level but underfits the curvature.

## 7 Numerical Results

All experiments use the real SPY reference contract: a near-at-the-money option with $S_0 = 728.99$, $K = 729$, $T = 0.23$, $r = 3.66\%$, the parity-implied carry $q = -2.12\%$, and at-the-money implied volatility $\sigma = 16.65\%$. Table 2 collects the European call across the four numerical methods; all agree to within Monte Carlo error, validating the implementations against one another. Table 3 reports the advanced-model cross-checks, each priced two independent ways on the same contract.

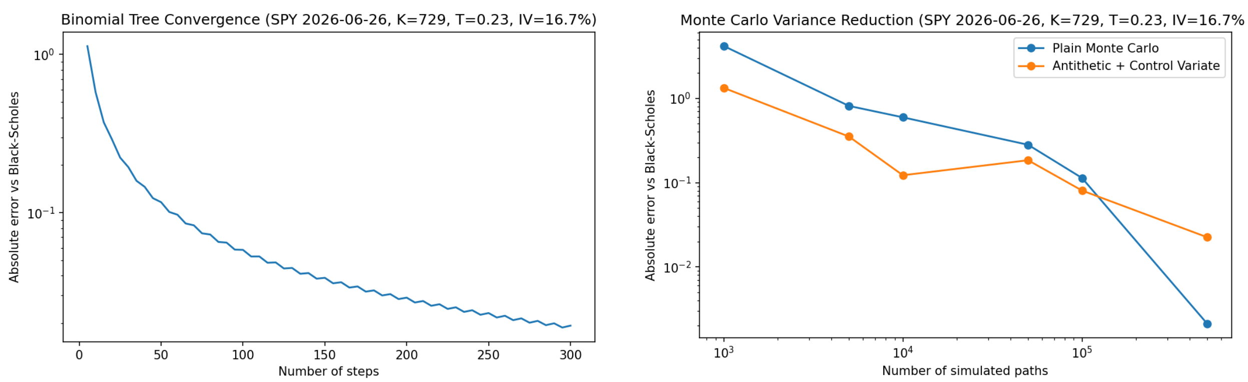
Table 2: European call on the real SPY contract ($S_0 = 728.99$, $K = 729$, $T = 0.23$, $r = 3.66\%$, $q = -2.12\%$, $\sigma = 16.65\%$). "SE" is the Monte Carlo standard error.

| Method | Price | Note |
|---|---|---|
| Black–Scholes (closed form) | 28.334 | benchmark |
| Binomial tree (500 steps) | 28.322 | $O(1/N)$ convergence |
| Monte Carlo (200k paths) | 28.338 | SE = 0.038 |
| Quasi-Monte Carlo (Sobol, $2^{16}$) | 28.333 | error $7 \times 10^{-4}$ |

Table 3: Cross-validation of the advanced models on the SPY contract using the calibrated parameters. Each row prices the same contract two independent ways; "SE" is the Monte Carlo standard error. Prices differ across rows because each model is priced with its own calibrated (full-smile) parameters, not the single at-the-money volatility.

| Model | Method A | Method B | A | B |
|---|---|---|---|---|
| Heston call | Fourier | Monte Carlo | 28.545 | $28.732 \pm 0.067$ |
| Merton call | Series | Monte Carlo | 27.990 | $27.996 \pm 0.045$ |
| American put | Lattice | LSM | 19.500 | $19.505 \pm 0.077$ |

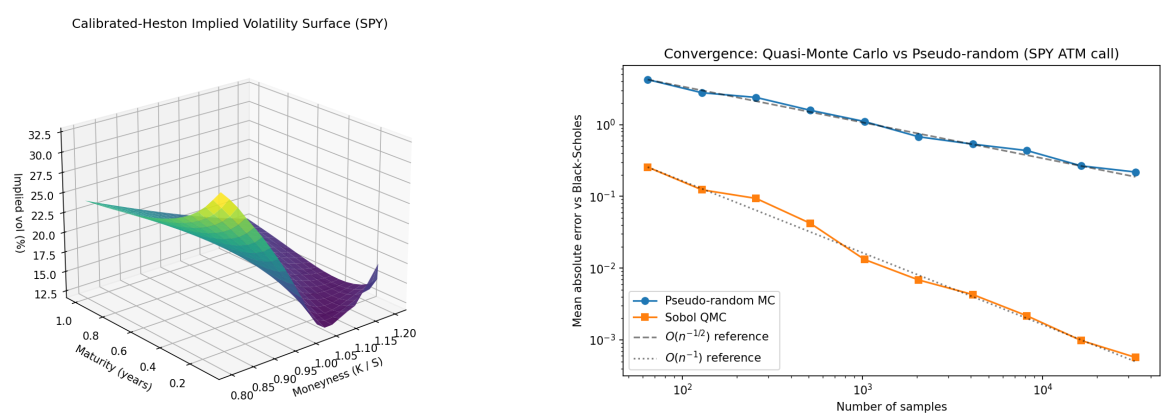
The variance-reduction and convergence experiments are summarised in the figures, all computed on the real SPY contract. Figure 7a shows the binomial tree converging to Black–Scholes. Figure 7b shows antithetic-plus-control-variate Monte Carlo dominating plain Monte Carlo at every sample size. Figure 8b shows the quasi-Monte Carlo convergence-rate improvement. The market smile, calibrated surface, and calibration-fit figures appear in Sections 4 and 6.




(a) Binomial tree convergence to BSM.

(b) Monte Carlo variance reduction.

Figure 7: Convergence and variance reduction for the European benchmark.




(a) Calibrated-Heston implied-volatility surface.

(b) QMC vs. pseudo-random convergence.

Figure 8: Left: the SPY-calibrated Heston surface—skew steepest at short maturities, flattening with $T$. Right: Sobol QMC error tracks $O(n^{-1})$ while pseudo-random MC tracks $O(n^{-1/2})$.

## 8 Conclusion

We have built and empirically validated a hierarchy of option-pricing models and numerical methods, organised around the failure of the Black–Scholes constant-volatility assumption, and applied to real S&P 500 option data. The Heston and Merton models reproduce the market volatility skew and its term structure that Black–Scholes cannot; semi-analytic Fourier pricing, variance-reduced and quasi-Monte Carlo simulation, and regression-based American exercise make these models computationally tractable; and calibration to live quotes closes the loop from market data back to model parameters. The recurring theme is cross-validation: each price is confirmed by an independent method, which is both a correctness discipline and, for a practitioner, a source of confidence in the numbers.

The principal quantitative findings, all on real SPY data, are:

- Calibrated to 903 liquid SPY quotes across six maturities, Heston fits the market skew to 0.60% implied-volatility RMSE and Merton to 1.22% (Table 1, Figure 6); the estimated $\rho = -0.71$ and mean jump $\mu_J = -0.13$ quantify the market's leverage effect and crash-risk pricing.

- All four European methods agree on the real SPY call to within Monte Carlo error (Table 2); the Heston Fourier price, Merton series, and American LSM price each match their independent counterpart to within one to three standard errors (Table 3).

- The correlation $\rho$ governs the skew and the vol-of-vol $\xi$ the convexity of the Heston smile (Figure 3), and the calibrated models' non-Gaussian return distributions (Figure 2) are the mechanism that generates the observed skew.

- Variance reduction is highly effective: a geometric-Asian control variate cuts the arithmetic-Asian standard error by roughly 36×, and Sobol QMC improves the European convergence rate from $O(n^{-1/2})$ toward $O(n^{-1})$ (Figure 8b).

Natural extensions include the Bates model (Heston with jumps), the SABR model for interest-rate smiles, Andersen's Quadratic-Exponential scheme for unbiased Heston simulation, stochastic local volatility for exact surface fitting, and adjoint algorithmic differentiation for fast Greeks. All code, tests, and figure-generating scripts are available in the companion repository.

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